1. Create Dataset in Databricks

```
from pyspark.sql.functions import col

appointments_data = [
    (1001, "Hyderabad", "Cardiology", "Apollo", 1500, "Completed"),
    (1002, "Bangalore", "Neurology", "Yashoda", 2200, "Completed"),
    (1003, "Mumbai", "Dermatology", "Care", 900, "Pending"),
    (1004, "Delhi", "Orthopedics", "Max", 2500, "Completed"),
    (1005, "Chennai", "Pediatrics", "Apollo", 1200, "Cancelled"),
    (1006, "Hyderabad", "Cardiology", "Care", 3000, "Completed"),
    (1007, "Bangalore", "Dermatology", "Apollo", 1000, "Completed"),
    (1008, "Mumbai", "Neurology", "Max", 2600, "Pending"),
    (1009, "Delhi", "Cardiology", "Yashoda", 2800, "Completed"),
    (1010, "Chennai", "Orthopedics", "Care", 2400, "Completed"),
    (1011, "Hyderabad", "Pediatrics", "Apollo", 1100, "Completed"),
    (1012, "Bangalore", "Cardiology", "Max", 3200, "Completed"),
    (1013, "Mumbai", "Pediatrics", "Yashoda", 1300, "Cancelled"),
    (1014, "Delhi", "Neurology", "Apollo", 2700, "Completed"),
    (1015, "Chennai", "Dermatology", "Care", 950, "Pending")
]

columns = [
    "appointment_id",
    "city",
    "department",
    "hospital",
    "consultation_fee",
    "status"
]

df = spark.createDataFrame(appointments_data, columns)

display(df)
```

Exercises

2. Bar Chart: Revenue by City

3. Bar Chart: Revenue by Department

4. Pie Chart: Appointment Status

5. Horizontal Bar: Revenue by Hospital

6. Scatter Plot: Appointment ID vs Fee

7. Line Chart: Fee Trend by Appointment

```
1. Create bar chart for appointment count by city.
2. Create bar chart for appointment count by department.
3. Create pie chart for status distribution.
4. Create bar chart for revenue by city.
5. Create bar chart for revenue by department.
6. Create horizontal bar chart for revenue by hospital.
7. Create line chart for consultation fee trend.
8. Create scatter plot using appointment_id and consultation_fee.
9. Create chart only for Completed appointments.
10. Create revenue chart after excluding Cancelled appointments.
11. Rotate x-axis labels by 45 degrees.
12. Change chart title.
13. Change x-axis label.
14. Change y-axis label.
15. Increase figure size.
16. Display top 3 cities by revenue.
17. Display top 3 departments by revenue.
18. Display lowest revenue hospital.
19. Create pie chart showing hospital-wise appointment share.
20. Create final EDA dashboard cells with 4 separate charts.
```